

Healthcare Support Center

Performance Analytics Dashboard

Power BI Case Study

Executive Summary

This case study demonstrates a comprehensive dual-dashboard analytics solution developed for a healthcare support organization operating 10 regional centers. The project encompassed two key deliverables: (1) Performance Analytics tracking visit KPIs against targets, and (2) CRM Integration Analysis identifying data quality issues across two separate systems.

Business Impact	Enabled monitoring of 5 strategic KPIs across 10 centers with real-time target tracking
Technical Scope	13 data tables, 11 relationships, 38 DAX measures, dynamic parameter-driven analysis
Data Quality Impact	Identified 32% contact overlap across CRM systems contributing 25% of total income

Business Challenge

The Problem

The organization faced critical operational and data management challenges:

- No unified view of performance across 10 regional centers making strategic planning difficult
- Limited ability to track actual performance against established targets across 5 KPIs
- Two separate CRM systems operating independently with unknown data overlap
- Potential duplicate contacts inflating metrics and skewing revenue attribution
- Manual Excel-based reporting consuming significant staff time with limited insights

Business Requirements

Leadership needed dashboards that would:

- Track total visits, new person with cancer visits, new family and friend visits, male:female ratio, and new:return visit ratio
- Compare 2024 performance against targets at both organizational and center levels
- Identify underperforming centers requiring intervention and high performers for best practice sharing
- Enable flexible analysis through user-controlled metric selection
- Quantify CRM system overlap and understand cross-system contact behavior

Solution Overview

Task 1: Performance Analytics Dashboard

Built a three-page performance monitoring system with the following components:

Page 1: KPI Performance Overview

- Five KPI cards showing organizational totals: Total Visits (330K+), New PWC (47K), New F&F (25K), Gender Ratio (53%), New:Return Ratio (29%)
- Bar chart breaking down visits by category showing 212K Person with Cancer vs 73K Family & Friend visits
- Line chart showing visit trends over time from January 2023 to July 2024
- Doughnut charts visualizing gender distribution (65% Female, 35% Male) and visit type breakdown (74% PWC, 26% F&F)
- Comprehensive matrix table displaying all metrics by center with performance percentages ranging from 88% to 110%

Page 2: Target Performance Analysis

- Dynamic metric selector allowing users to choose from 5 KPIs with instant chart updates
- Stacked bar chart showing actual (beige) vs target (dark red/orange) for selected metric across all 10 centers
- Waterfall chart visualizing net performance difference showing total variance of £12,324 above target
- Detailed table with actual, target, and variance calculations with conditional formatting (green for positive, red for negative)

Page 3: Dynamic Scatter Analysis

- Interactive bubble chart with user-controlled X-axis (New Person with Cancer Visits), Y-axis (Family & Friend Visits), and size dimension (Return PWC Visits)
- Three parameter slicers enabling exploration of any metric combination for correlation analysis
- Quadrant analysis showing center positioning across performance dimensions
- Supporting table showing selected metrics for each center with totals

Task 2: CRM Integration Analysis

Developed a two-page cross-system analysis dashboard:

Page 1: System A Overview

- Four KPI cards: Total Contacts (50), Activities (300), Unique Contacts (50), Cross-CRM Count (16 representing 32%)
- Doughnut charts showing activity breakdown by type (Workshop 30%, Session 30%, Group 40%), geographic distribution (England 54%, Scotland 20%, Wales 25%), and gender split (Male 44%, Female 44%, Blank 12%)
- Bar charts analyzing contacts by age group (40-65 highest at 17) and contact type (Person with Cancer dominant at 20)
- Time series trend line showing activity volume fluctuations throughout 2024

Page 2: System B Income Analysis

- Five income KPIs: B-CRM Contacts (48), Activities (150), Total Income (£1,688), Cross-CRM Income (£423), Cross-CRM % (25%)
- Waterfall chart breaking down income by type: Donation (£513), Other (£442), Sponsor (£379), Event Ticket (£354)
- Line chart showing income trends with notable spikes in March and July 2024
- Matrix table cross-referencing income type by cross-CRM status showing Donations contribute 30.4% and cross-CRM contacts generate 25.1% of total income

Technical Implementation

Data Preparation & Transformation

CentreActivity Table

- Renamed columns for clarity: 'NewOrReturning' to 'VisitType', 'DeliveryDate' to 'Date'
- Filtered to only 'Present' attendance status and valid genders (Man, Woman)
- Created calculated column 'Visit Category' grouping ContactType into 'Person with Cancer' or 'Family and Friend'
- Extracted time dimensions: Year, Month Name, Month Sort, Quarter, Quarter Sort, Day Name, Day Sort

CentreTargets Table (Critical Transformation)

The source data had years as column names (TotalVisitTarget2023, TotalVisitTarget2024) requiring restructuring:

- Transposed data in Excel to move years from columns into rows, creating proper dimensional structure
- Split ratio columns (NewToReturnVisitPWCTarget: '3:7') into numerators and denominators, then calculated decimal values
- Final schema: Centre, Year, TotalVisitTarget, NewPersonWithCancerTarget, NewFamilyandFriend Target, NewToReturnVisitPWCTarget, MaleToFemaleVisitTarget

CRM Contact Matching Logic

Implemented fuzzy matching to identify contacts existing in both systems:

1. Removed system prefixes ('A-', 'B-') from contact IDs and created sort columns
2. Standardized emails (lowercase, trimmed whitespace)
3. Merged A-CRM Contacts with B-CRM Contacts on email to flag cross-system matches
4. Created 'Cross-CRM Contact' column with values 'Cross CRM' or 'Not Cross CRM'
5. Linked B-CRM Income to A-CRM Contacts for revenue attribution across systems

Data Model Architecture

Task 1 Model

- CentreActivity → DimCentre (Many-to-One on Centre)
- CentreTargets → DimCentre (Many-to-One on Centre)
- CentreActivity → CentreTargets (Many-to-Many on Centre and Year with bidirectional filtering)
- Disconnected parameter tables: Selection, X_Axis, Y_Axis, Bin_Size

Task 2 Model

- A-CRM Activity → A-CRM Contacts (Many-to-One on ContactId)
- B-CRM Income → B-CRM Contacts (Many-to-One on ContactId)
- A-CRM Contacts → B-CRM Contacts (Many-to-One on matched B-CRM Contact ID)
- B-CRM Income → A-CRM Contacts (Many-to-Many for revenue attribution)
- Date dimension (Dim date) linked to both A-CRM Activity and B-CRM Income

DAX Measures Examples

Developed 38 measures across both dashboards. Key examples:

Visit Performance Calculation

Visit Performance (Actual Vs Target) = DIVIDE([Total Visits], [Total Visit Target], 0)

Safe division returning 0 when target is missing. Formatted as percentage for immediate performance assessment.

Cross-System Contact Identification

Cross CRM = CALCULATE([Total A-CRM Contacts], 'A-CRM Contacts'[Cross-CRM Contact] = "Cross CRM")

Filters total contacts to only those flagged as existing in both CRM systems through the matching transformation.

Revenue Attribution

Cross CRM Income % = DIVIDE([Cross CRM Income], [Total Income], 0)

Calculates percentage of total income generated by contacts present in both systems.

Dynamic User Selection (Advanced)

Dynamic Selected Actual = SWITCH(SELECTEDVALUE(Selection[Selector]), "Total Visit", [Total Visits], "New PWC", [New Person with Cancer Visits], ...)

Enables user-driven metric switching in visuals by evaluating slicer selection and returning corresponding measure. Powers the dynamic target analysis page.

Key Findings & Insights

Task 1: Performance Insights

- Centre10 achieved highest performance with 26,113 visits (106% of target), exceeding goal by 1,113 visits
- Centre7 also exceeded target with 23,932 visits (102%), showing strong community engagement
- Centre4 and Centre8 underperformed, falling short by 875 and 1,044 visits respectively, requiring intervention
- Overall New:Return PWC ratio at 29.23% suggests opportunities to improve conversion of new visitors to returning visitors
- Gender imbalance with 65% female visitors (215,534) vs 35% male (115,252) indicates need for targeted male outreach
- Organization exceeded total target by 12,324 visits (107.7% achievement), demonstrating strong overall performance

Task 2: CRM Integration Insights

- 32% of System A contacts (16 out of 50) also exist in System B, indicating significant data duplication
- Cross-CRM contacts contribute 25% (£423) of total income (£1,688) despite representing only 32% of unique contacts
- England centers show strongest activity at 54% (163 activities), followed by Wales at 25% and Scotland at 20%
- Group activities (40%) and Workshop activities (30%) dominate, with Sessions at 30%
- Donations drive 30.4% of total income (£513), making it the largest revenue stream
- Cross-CRM contacts contribute consistently across all income types, with particularly strong representation in Donations (4.8% from cross-CRM)
- Age group 40-65 is most represented with 17 contacts, followed by 21-40 with 13 contacts

Strategic Recommendations

Performance Optimization

6. Deploy intervention resources to Centre4 and Centre8 to address underperformance (875 and 1,044 visits below target respectively)
7. Document and replicate Centre10 and Centre7 best practices across the organization (both exceeded targets by 100+ visits)
8. Launch targeted male outreach initiatives to balance 65:35 gender ratio, particularly in centers showing higher disparities
9. Develop strategies to improve 29% New:Return PWC ratio through enhanced follow-up programs and engagement initiatives

CRM System Improvements

10. Prioritize cross-CRM contacts for fundraising campaigns given their 25% income contribution despite representing only 32% of contacts
11. Implement master data management to resolve duplicate contacts and improve reporting accuracy across both systems
12. Evaluate CRM consolidation to single platform, reducing duplication and improving data quality
13. Leverage England centers (54% of activity) to pilot new programs before organization-wide rollout
14. Focus donation campaigns on proven success areas, as Donations contribute 30.4% of income
15. Target 40-65 age group (highest representation) for specialized programming and engagement

Technical Skills Demonstrated

Skill Category	Specific Competencies
Data Preparation	Data transposition, column splitting, fuzzy matching, data standardization
Data Modeling	Star schema, many-to-many relationships, bidirectional filtering, parameter tables
DAX	CALCULATE, DIVIDE, SWITCH, SELECTEDVALUE, filter context manipulation
Visualization	KPI cards, matrix tables, scatter plots, waterfall charts, conditional formatting
Business Analysis	KPI development, target tracking, performance benchmarking, data quality assessment

Conclusion

This dual-dashboard solution demonstrates end-to-end analytics capabilities from complex data preparation through actionable visualization design. The project required solving real-world data challenges including dimensional restructuring, cross-system matching, and dynamic parameter implementation.

By combining technical proficiency in Power BI with strong business acumen, the solution delivers immediate operational value through performance monitoring while identifying strategic opportunities through CRM integration analysis. The dashboards enable data-driven decision-making at both tactical and strategic levels, with clear actionable insights derived from 330,000+ visit records and £1,688 in revenue data.